



Dashboard:

[Live Streamlit Dashboard](#)



BIRD SPECIES OBSERVATION ANALYSIS

Forest & Grassland Habitat Monitoring

Data Analytics Project

Tools: Python • Pandas • SQLite • Streamlit • Plotly

Kartikey Singh

Data Analyst | Business Analyst

Bird Species Observation Analysis

I. Project Overview

This project analyzes bird observation data collected across **Forest** and **Grassland** habitats to identify patterns in species distribution, habitat activity, observation timing, environmental conditions, detection methods, survey effort, and conservation significance.

The raw monitoring data was collected across multiple observation sheets and required integration, cleaning, validation, and transformation before analysis. Forest and Grassland observations were consolidated into a single analytical dataset, followed by exploratory and comparative analysis using Python and Pandas.

The cleaned dataset was subsequently stored in a **SQLite database** and used as the data source for an interactive **Streamlit dashboard**, allowing users to explore observations through habitat, date, species, plot, identification method, and conservation-status filters.

The overall objective was to transform raw bird-monitoring records into **clear, data-driven insights that can support habitat monitoring, biodiversity assessment, and conservation analysis**.

2. Project Objective

The analysis was designed to answer the following key questions:

- How are bird observations distributed between Forest and Grassland habitats?
- Which species are most frequently observed?
- How does observation activity vary across months and time of day?
- Which plots show the highest observation activity?
- What environmental conditions are associated with recorded observations?
- How do identification methods, observation distance, and flyovers differ between habitats?
- How does survey effort influence recorded observation activity?
- Which species and observations have conservation significance?

The analysis focuses on **recorded observations and survey activity**, rather than estimating bird population size.

3. Data & Preparation

3.1 Data Integration

The Forest and Grassland monitoring datasets were consolidated into a single analytical dataset. Individual observation sheets were combined within each habitat, and a Habitat field was added to distinguish Forest and Grassland records.

The integrated dataset provided a consistent structure for comparing observations across habitats while retaining information related to species, location, observation timing, environmental conditions, survey effort, detection methods, and conservation status.

3.2 Data Cleaning & Preprocessing

The dataset was prepared for analysis through several cleaning and transformation steps:

- Converted Year and Visit into appropriate numeric formats.
- Converted Start_Time and End_Time into datetime formats.
- Standardized missing categorical values.
- Replaced missing ID_Method, Distance, and Sex values with Not Recorded.
- Checked for exact and potential duplicate observations.
- Removed exact duplicate records.
- Created derived temporal fields including Observation_Hour, Time_Period, Month_Number, and Month_Name.

3.3 Data Quality Validation

Data quality was evaluated after cleaning through:

- Missing-value analysis
- Not Recorded value checks
- Exact duplicate verification
- Completeness checks for key analytical fields
- Validation of important categorical and numerical variables

These steps ensured that the final dataset was suitable for exploratory analysis and dashboard visualization.

4. Analytical Approach

The analysis followed a structured workflow:

Raw Data → Data Integration → Data Cleaning & Preprocessing → Data Quality Validation → Exploratory Analysis → Habitat Comparison → Temporal Analysis → Spatial Analysis → Species Analysis → Environmental Analysis → Observer & Visit Context → Distance & Behaviour → Conservation Analysis → Dashboard

Habitat Analysis

Observation volume, species richness, species overlap, and plot-level activity were analyzed across Forest and Grassland habitats.

Temporal Analysis

Observation patterns were analyzed across months, observation hours, and broader time periods to identify periods of higher recorded activity.

Species Analysis

Species were ranked by observation frequency and examined across habitats to identify dominant, shared, and habitat-specific species.

Spatial & Survey Effort Analysis

Observation activity was analyzed at plot level, including survey sessions and observations per session, to provide context for differences in recorded activity.

Environmental & Detection Analysis

Temperature, humidity, sky, wind, disturbance, identification method, observation distance, and flyover observations were analyzed to understand differences in observation conditions and detection patterns.

Conservation Analysis

PIF Watchlist and Regional Stewardship indicators were analyzed to identify observations and species with greater conservation relevance.

5. Dashboard & Visualization

An interactive **Streamlit dashboard** was developed to provide a user-friendly interface for exploring the bird observation dataset.

The dashboard allows users to filter observations by:

- Date range
- Habitat
- Month
- Species
- Plot
- Identification method
- Conservation status

The dashboard is organized into four analytical areas:

- **Overview**
- **Species**
- **Environment & Detection**
- **Conservation & Plots**

The dashboard connects to the SQLite database and loads the observation records into Pandas for visualization and interactive analysis.

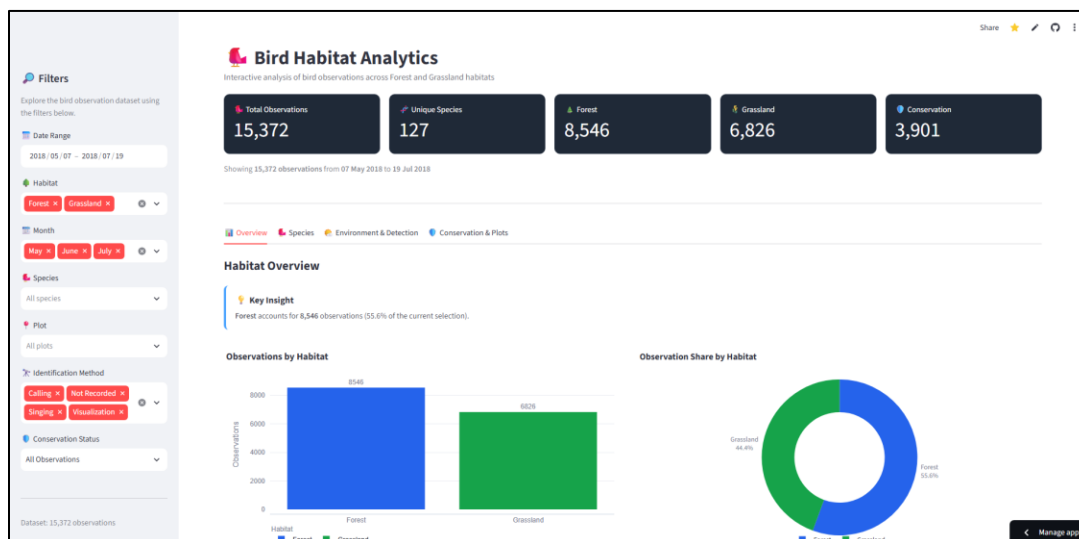


Figure 1. Streamlit dashboard overview showing total observations, species count, and habitat-level observation volume.

6. Key Findings

The analysis identified several notable patterns across the two habitats:

- **Forest recorded 8,546 observations**, compared with **6,826 in Grassland**.
- Grassland had higher plot-level activity, with **33.96 observations per plot** compared with **20.95 in Forest**.
- Species richness was very similar, with **108 species in Forest** and **107 in Grassland**, including **88 shared species**.
- **June was the peak month for Forest**, while Grassland observations were distributed more evenly across May, June, and July.
- **7 AM was the peak observation hour** in both habitats.
- Most observations occurred within the **20–25°C** temperature band.
- **Singing** was the dominant identification method in both habitats.
- Grassland had substantially higher proportions of **visual identifications and flyover observations**.
- Forest had a considerably **higher share of conservation-status observations**, particularly PIF Watchlist observations.
- **ANTI-0163 in Grassland recorded the highest observation count for a single plot, with 54 observations.**

7. Actionable Insights

The findings provide several practical considerations for future monitoring and conservation planning:

1. Consider survey effort when evaluating habitat activity

Forest recorded more observations overall, but Grassland showed substantially higher observations per plot. Habitat comparisons should therefore consider **plot coverage and survey effort**, rather than relying only on total observation counts.

2. Prioritize high-activity plots for monitoring

High-activity plots such as **ANTI-0163** can be useful locations for continued monitoring and more detailed investigation of species activity and habitat characteristics.

3. Focus observation efforts around peak activity periods

Because recorded observations peaked around **7 AM**, early-morning survey periods may provide greater observation activity and can be considered when planning future monitoring sessions.

4. Consider conservation-status observations in habitat planning

Forest showed a considerably higher proportion of PIF Watchlist and Regional Stewardship observations. These observations can help identify areas where **additional conservation attention and monitoring may be valuable**.

5. Account for detection differences

The higher proportion of visual identifications and flyovers in Grassland indicates that observation profiles differ between habitats. Detection method and observation distance should therefore be considered when interpreting differences in recorded activity.

8. Data Limitations

The analysis is based on **recorded observations**, which reflect both bird activity and survey effort.

Several limitations should therefore be considered:

- Higher observation counts do not necessarily indicate larger bird populations.
- Forest and Grassland have different numbers of monitoring plots, affecting total observation volume.
- Differences in survey effort may influence habitat-level observation counts.
- Identification method and observation distance can influence the probability of recording an observation.
- Environmental results represent observed associations and should not be interpreted as causal relationships.
- Conservation-status observations indicate recorded observations associated with the relevant status categories and do not represent population estimates.

These limitations should be considered when using the findings for ecological or conservation decisions.

9. Conclusion

This project transformed raw bird-monitoring records from Forest and Grassland habitats into a structured, validated, and interactive analytical solution.

The analysis showed that **Forest contributed greater overall observation volume and a higher share of conservation-status observations, while Grassland demonstrated greater observation activity per plot and higher proportions of visual and flyover observations.**

Despite differences in observation volume and activity patterns, the two habitats showed **very similar species richness**, with a substantial number of species shared between them.

The combination of Python-based analysis, SQLite data storage, and an interactive Streamlit dashboard provides a practical framework for exploring bird observations and supporting **habitat monitoring, biodiversity assessment, and conservation-focused decision-making.**